1.5.6 Distributive property/law for the collection $\mathbb{Q}$ of rational numbers

Multiplication is distributive over addition for the collection of rational numbers.

For any three rational numbers ***a***, ***b*** and ***c***, the distributive law is $a\times(b+c)=(a\times b)+(a\times c)$

Illustration

Take rational numbers a,b,c as $a=\frac{-7}{9}$, $b=\frac{11}{18}$ and $c=\frac{-14}{27}$

$$\text{Now, } b+c = b+c=\frac{11}{18}+\frac{-14}{27}=\frac{33}{54}+\frac{-28}{54}=\frac{33-28}{54}=\frac{5}{54}$$

(equivalent rationals with common denominators)

$$\therefore\ a\times(b+c) = \frac{-7}{9}\times\frac{5}{54}=\frac{-7\times5}{9\times54}=\frac{-35}{486} \qquad \ldots(1)$$

$$\text{Also, } a\times b = \frac{-7}{9}\times\frac{11}{18}=\frac{-7\times11}{9\times18}=\frac{-77}{9\times9\times2}$$

$$a\times c = \frac{-7}{9}\times\frac{-14}{27}=\frac{7\times14}{9\times9\times3}=\frac{98}{9\times9\times3}$$

$$\therefore\ (a\times b)+(a\times c) = \frac{-77}{9\times9\times2}+\frac{98}{9\times9\times3}$$

$$=\frac{-77\times3+98\times2}{9\times9\times2\times3} = \frac{-231+196}{486}=\frac{-35}{486} \qquad \ldots(2)$$

(1) and (2) shows that $a\times(b+c)=(a\times b)+(a\times c)$.

Hence, multiplication is distributive over addition for the collection $\mathbb{Q}$ of rational numbers.

Exercise 1.3

1. Verify the closure property for addition and multiplication for the rational numbers $\frac{-5}{7}$ and $\frac{8}{9}$.
2. Verify the commutative property for addition and multiplication for the rational numbers $\frac{-10}{11}$ and $\frac{-8}{33}$.
3. Verify the associative property for addition and multiplication for the rational numbers $\frac{-7}{9},\frac{5}{6}$ and $\frac{-4}{3}$.
4. Verify the distributive property $a\times(b+c)=(a\times b)+(a+c)$ for the rational numbers $a=\frac{-1}{2}$, $b=\frac{2}{3}$ and $c=\frac{-5}{6}$.
5. Verify the identity property for addition and multiplication for the rational numbers $\frac{15}{19}$ and $\frac{-18}{25}$.

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